

Corrigé exercice 38 :

1. $\bar{z} = \overline{(1+i)^2} = (1-i)^2 = 1 - 1 - 2i = -2i$

2. $\bar{z} = \overline{(2+i)^3} = (2-i)^3 = 2^3 - 3 \times 2^2 \times i + 3 \times 2 \times i^2 - i^3 = 8 - 12i - 6 + i = 2 - 11i$

3. $\bar{z} = \overline{(1-i)^4} = (1+i)^4 = 1 + 4i + 6i^2 + 4i^3 + i^4 = 1 + 4i - 6 - 4i + 1 = -4$

4. $\bar{z} = \overline{(3+2i)^3} = (3-2i)^3 = 3^3 - 3 \times 3^2 \times 2i + 3 \times 3 \times (2i)^2 - (2i)^3 = -9 - 46i$